

Electronics Mini-Project 1

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1. Introduction

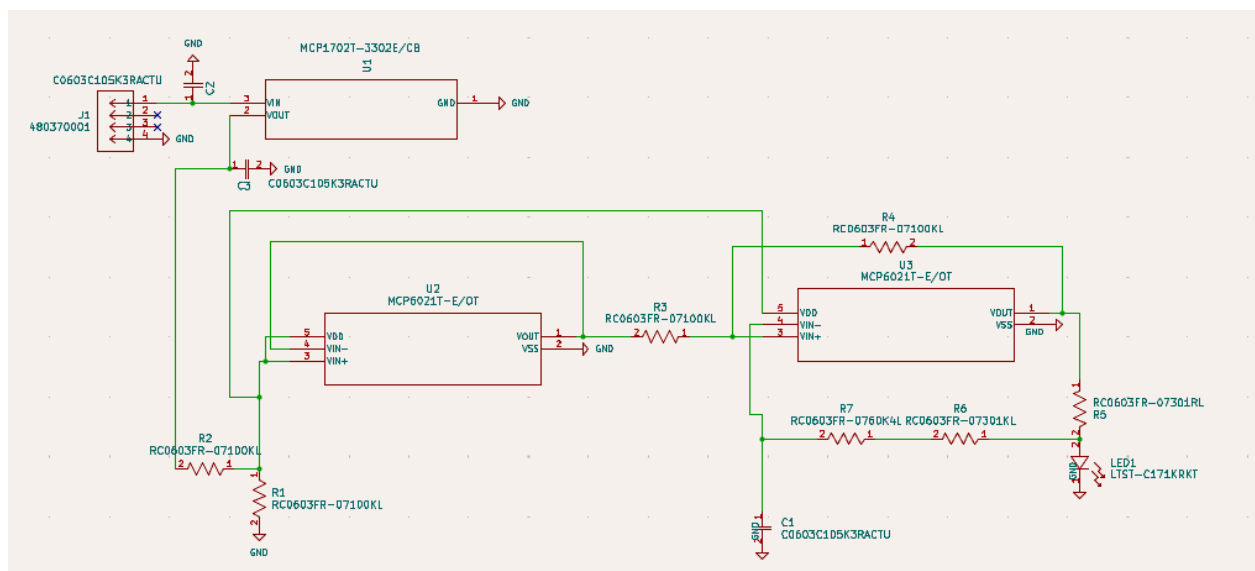
The goal of this project is to design a circuit that flashes an LED within a period of 1 second, with a total tolerance of $\pm 10\%$.

The circuit I made includes a unity-gain buffer, establishing a reference that is half of the input voltage, which feeds into another op-amp. This second op-amp, alongside a resistor-capacitor circuit that feeds into the comparator, allows for the output of the op-amp to oscillate, causing the LED to flash on and off.

In order to complete this project, I designed an initial circuit in LTspice and performed Monte Carlo Simulations. Then, the schematic was remade within KiCad 10.0 and turned into a final PCB design.

2. Circuit architecture

The circuit divides into four stages: The USB 5V VBUS input, the voltage converter, the unity-gain buffer, and the comparator op-amp, which feeds into the flashing LED.



3. Power supply

The board draws power from USB VBUS and regulates it down to 3.3 V with an MCP1702T-3302E/CB low-dropout regulator (U1 in the KiCad schematic).

The USB Type_A plug has 4 pins: pin 1 for VBUS and pin 4 for GND. Pins 2 and 3 are data pins that are unused in this project.

The voltage regulator also makes use of bypass capacitors. In the datasheet of the component, it is required to have a capacitor between 1.0 μ F and 22 μ F. These capacitors allow for a smoother voltage output that is more robust to voltage spikes and drops.

It is important to note that in the LTspice model, this stage is omitted and is instead replaced by an ideal 3.3V power supply. This does not impact the simulation, as the oscillation is independent of the voltage.

4. Reference and buffer stage

Coming from the voltage regulator, the voltage of the buffer stage is roughly 3.3V, but this goes through a voltage divider (R5 and R6), both 100 k Ω , that inputs into the unity-gain buffer. The divider ratio is $\frac{1}{2}$, so the input voltage into the unity-gain buffer is $3.3\text{V}/2 = \sim 1.65\text{ V}$.

In this circuit, the buffer is necessary because an ideal buffer always reproduces its input voltage at its output, regardless of what's connected downstream. Without it, the reference node would connect directly to the oscillating comparator through R1. Because the comparator's output swings between the two rails as it oscillates, that connection would pull current from the reference divider in a way that varies with the oscillation, causing the reference voltage itself to shift over the cycle and changing the period along with it.

The reason this buffer is a unity-gain buffer is that the voltage must stay the same, and a unity-gain buffer has a gain of 1 (meaning the input is the same as the output).

5. Comparator Op-amp

U3, an op-amp in the schematics, and its surrounding components form the oscillator, which drives the flashing of the LED light. The derivation below produces the period equation that drives every component choice.

R1 runs from the buffered 1.65 V reference to U3's non-inverting input. R2 runs from that same node to the output. The node is therefore a two-resistor divider between Vref and Vout:

$$V_+ = V_{ref} \cdot R_2 / (R_1 + R_2) + V_{out} \cdot R_1 / (R_1 + R_2)$$

With $R_1 = R_2 = 100 \text{ k}\Omega$, this simplifies to the average of the two:

$$V_+ = (V_{ref} + V_{out}) / 2$$

Because the MCP6021 output is rail-to-rail, V_{out} is either V_{DD} or 0. The threshold therefore takes two values:

$$V_{out} = V_{DD}$$

$$V_+ = (V_{DD}/2 + V_{DD}) / 2 = 3 \cdot V_{DD}/4 = 2.475 \text{ V}$$

$$V_{out} = 0 \rightarrow V_+ = (V_{DD}/2 + 0) / 2 = V_{DD}/4 = 0.825 \text{ V}$$

The positive feedback through R_2 means the switching threshold moves away from the capacitor voltage the instant the output flips, which sets the voltage window the capacitor must traverse.

R_3 connects V_{out} to the capacitor node, and C_1 connects that node to ground. The capacitor charges exponentially toward whatever the output currently is, with time constant:

$$\tau = R_3 \cdot C_1.$$

U_3 's inverting input monitors this node, and because the op-amp input draws negligible current, R_3 and C_1 are the only elements that set the rate.

If the output were just switched high. The capacitor sits at the lower threshold $V_{DD}/4$ and charges toward V_{DD} . It rises until it reaches the upper threshold $3 \cdot V_{DD}/4$, at which point the inverting input exceeds the non-inverting input and the output switches low. Using the RC Charging equation for a first-order RC circuit

$$t_{charge} = R_3 \cdot C_1 \cdot \ln[(V_{DD} - V_{DD}/4) / (V_{DD} - 3 \cdot V_{DD}/4)] = R_3 \cdot C_1 \cdot \ln(3)$$

The output is now at 0, and the capacitor discharges toward 0 from $3 \cdot V_{DD}/4$ until it falls to $V_{DD}/4$:

$$t_{discharge} = R_3 \cdot C_1 \cdot \ln[(3 \cdot V_{DD}/4) / (V_{DD}/4)] = R_3 \cdot C_1 \cdot \ln(3)$$

The two halves are equal, so the output is a symmetric square wave and the full period is:

$$T = 2 \cdot R_3 \cdot C_1 \cdot \ln(3) = 2.1972 \cdot R_3 \cdot C_1$$

It is important to notice that the VDD terms are dropped in the calculations, highlighting that the period is not dependent on voltage.

6. Component value selection

From Section 5, $T = 2 \cdot R3 \cdot C1 \cdot \ln(3)$. Setting $T = 1.000$ s:

$$R3 \cdot C1 = T / (2 \cdot \ln 3) = 1.000 / 2.1972 = 0.4551 \text{ s}$$

The period fixes only the product, so one value is free.

C1 = 1 μ F. Chosen first, and chosen as this capacitor, as the tolerance is 5%, whereas capacitors higher in capacitance have a tolerance of 10%. This allowed the flashing of the circuit to remain within the tolerance

R3 = 361 k Ω .

Not an E96 value, so it is built as a series pair:

$$301 \text{ k}\Omega + 60.4 \text{ k}\Omega = 361.4 \text{ k}\Omega$$

R1 = R2 = 100 k Ω . Only the ratio matters. Equal values give:

$$V+ = (V_{ref} + V_{out})/2 \rightarrow \text{thresholds at } VDD/4 = 0.825 \text{ V and } 3 \cdot VDD/4 = 2.475 \text{ V}$$

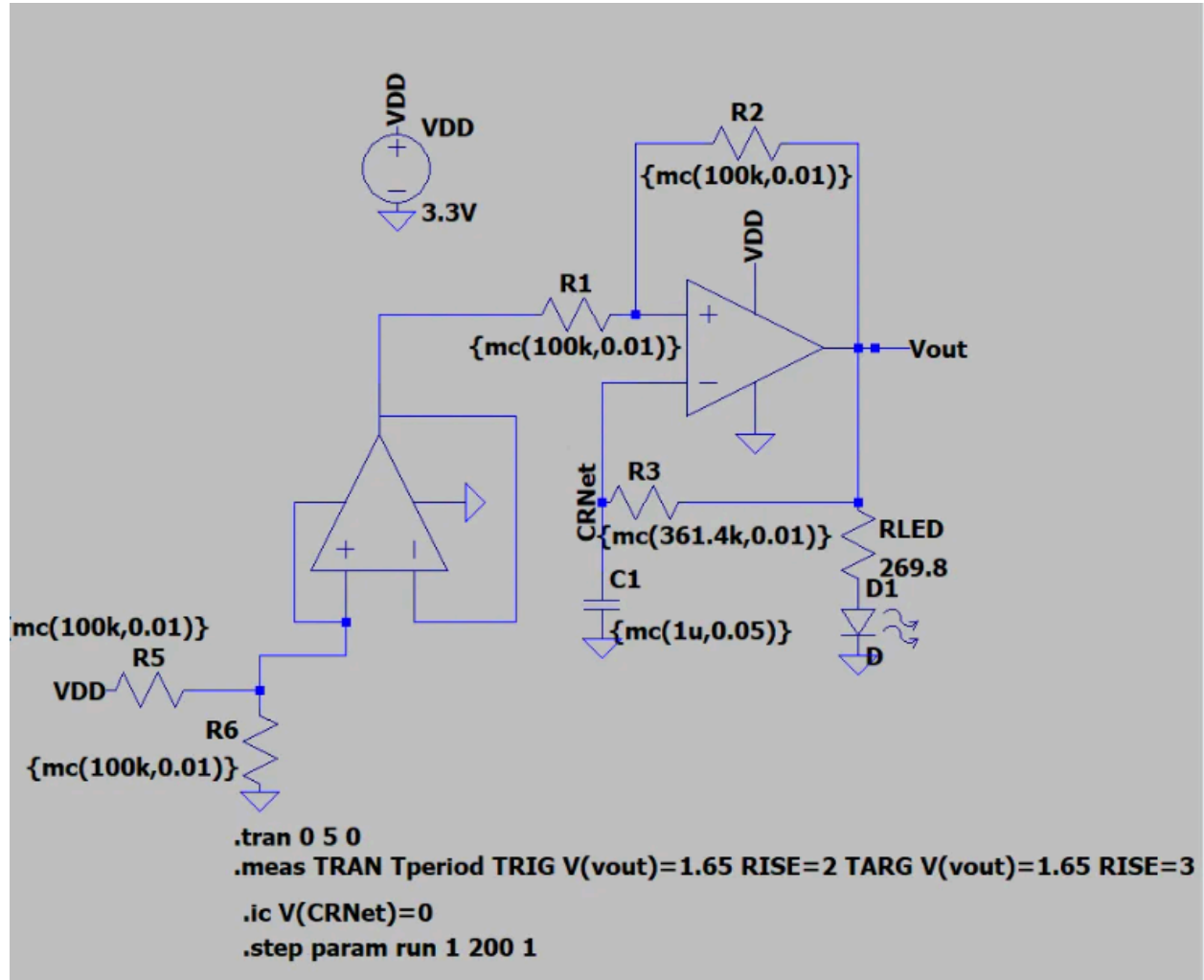
R5 = R6 = 100 k Ω . Sets $V_{ref} = VDD/2$ exactly.

RLED = 301 Ω . With $V_f = \sim 1.9$ V for the LTST-C171KRKT:

$$I_{LED} = (VDD - V_f) / R_{LED} = (3.3 - 1.9) / 301 = 4.65 \text{ mA}$$

Well inside the MCP6021's ~ 30 mA output capability, so the output still reaches its rails.

7. LTspice simulation setup



The LTspice model contains the reference divider, buffer, comparator, timing network, and LED branch, powered by an ideal 3.3 V source in place of the USB and regulator stage. Five directives control the simulation:

Directive

Function

`.tran 0 5 0`

Runs a transient analysis from $t = 0$ to $t = 5$ s with no suppressed interval, capturing roughly five full oscillation cycles per run.

```
.ic V(CRNet)=0
```

Sets the initial condition on the capacitor node to 0 V. Without it the solver picks its own starting point, and the first half-cycle would be an arbitrary length that corrupts the period measurement.

```
.meas TRAN  
Tperiod TRIG  
V(vout)=1.65  
RISE=1 TARG  
V(vout)=1.65  
RISE=2
```

Measures the period automatically. It triggers on the first rising crossing of 1.65 V and targets the second, reporting the elapsed time.

```
.step param run 1  
200 1
```

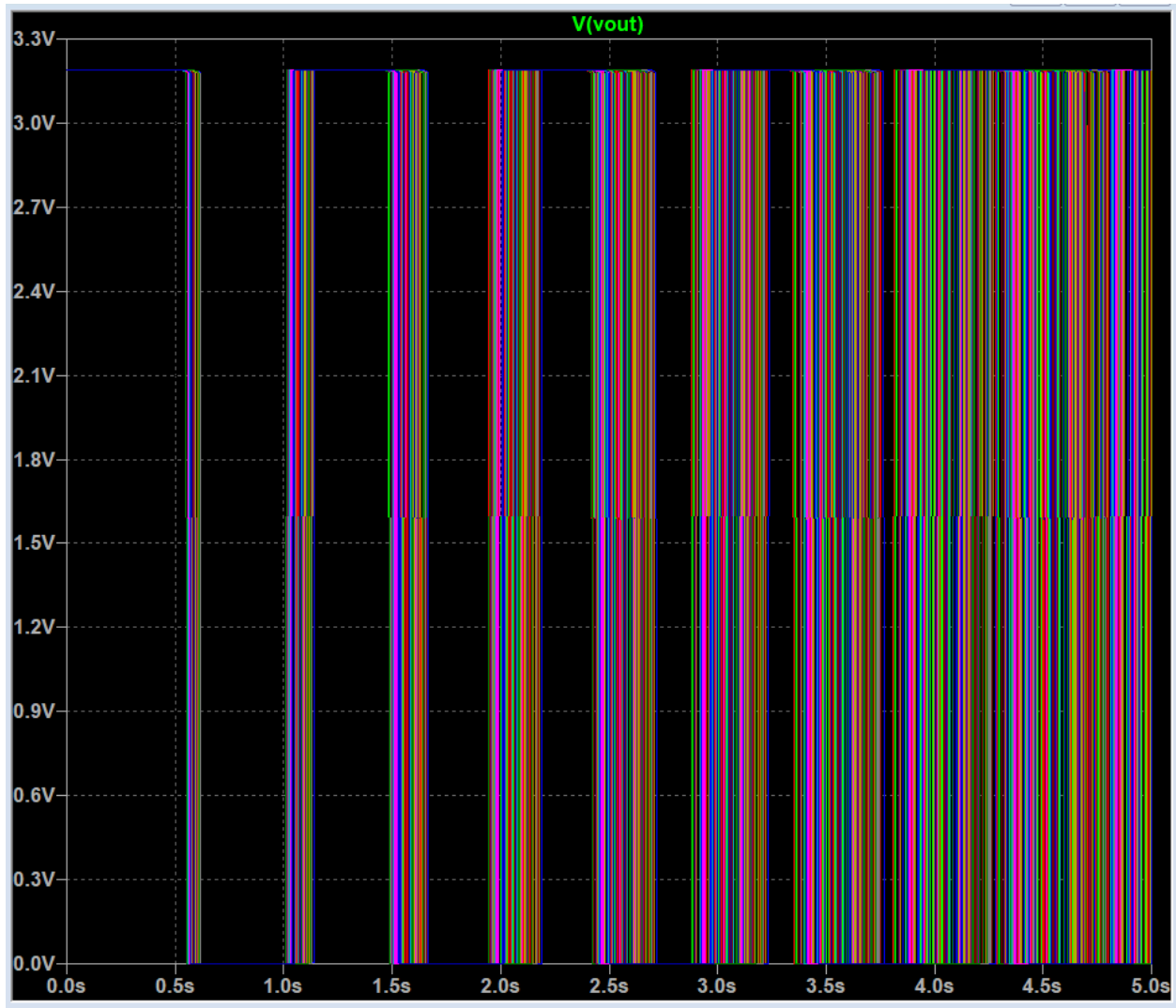
Sweeps a parameter from 1 to 200 in steps of 1, which executes the simulation 200 times.

```
{mc(nominal,  
tol)}
```

Wraps every passive value. On each run it draws a value from a uniform distribution within tolerance.

8. Tolerance analysis

Monte Carlo Simulations:

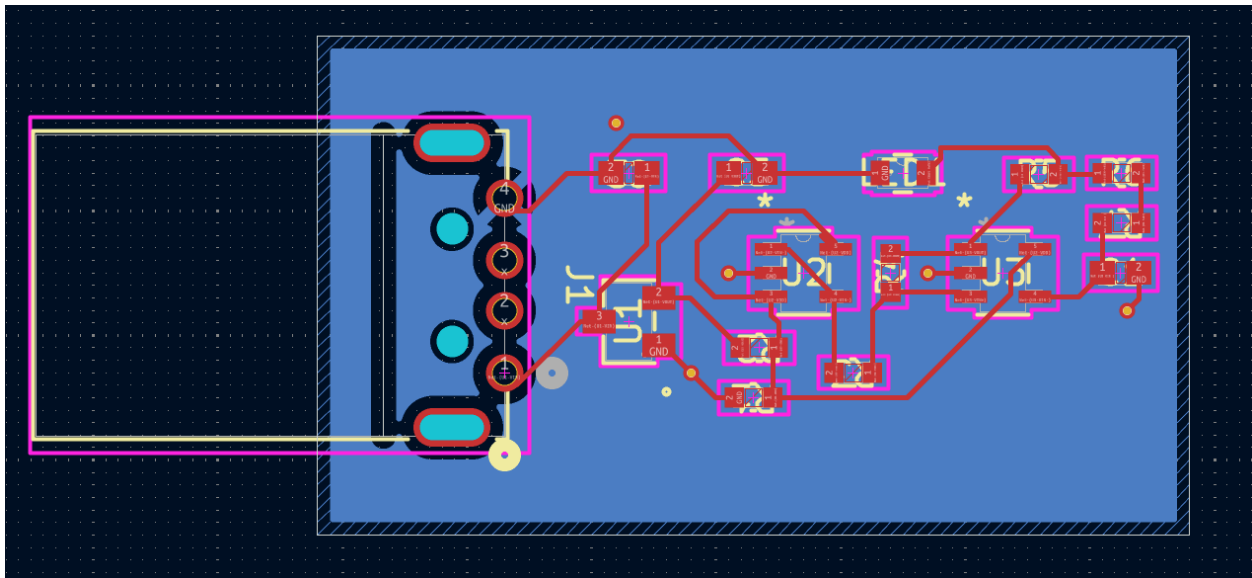
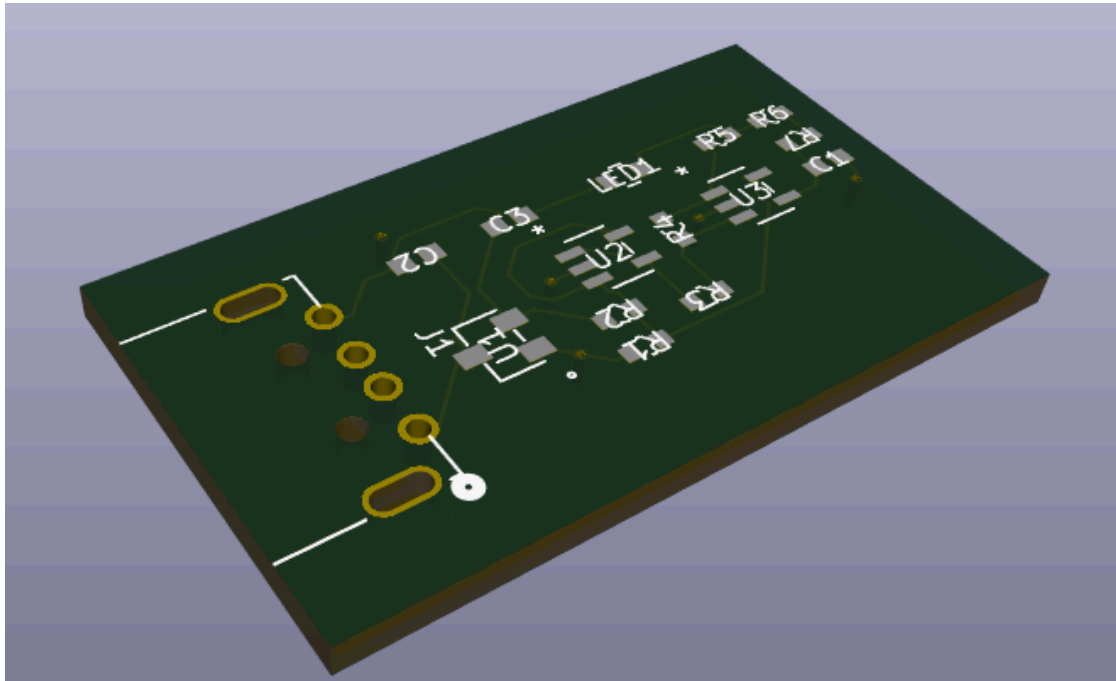


Results of VOut from 200 Monte Carlo Simulations, testing different tolerances of components

In each of the simulation results, the period landed within the range 0.9309 s to 1.0714 s, which fits the 10% tolerance.

9. PCB

The Size of the PCB is 35 mm by 20 mm.



10. Bill of materials

Qty	Value	Part number	Tolerance
1	—	Molex 0480370001	—
1	3.3 V	MCP1702T-3302E/C B	—
2	—	MCP6021T-E/OT	—
3	1 μ F	C0603C105K3RACT U	$\pm 5\%$
4	100 k Ω	RC0603FR-07100KL	$\pm 1\%$
1	301 k Ω	RC0603FR-07301KL	$\pm 1\%$
1	Red	LTST-C171KRKT	—